

Develop a Detection System of Deepfake Audio

深度偽造語音偵測系統

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Motivation and Purpose

Motivation

- Artificial intelligence is advancing rapidly, and many criminals are using it to commit fraud.
- Political Issues Related to Deepfake speech.
- Humans can only detect deepfake speech with an accuracy rate of 73%.

Motivation and Purpose

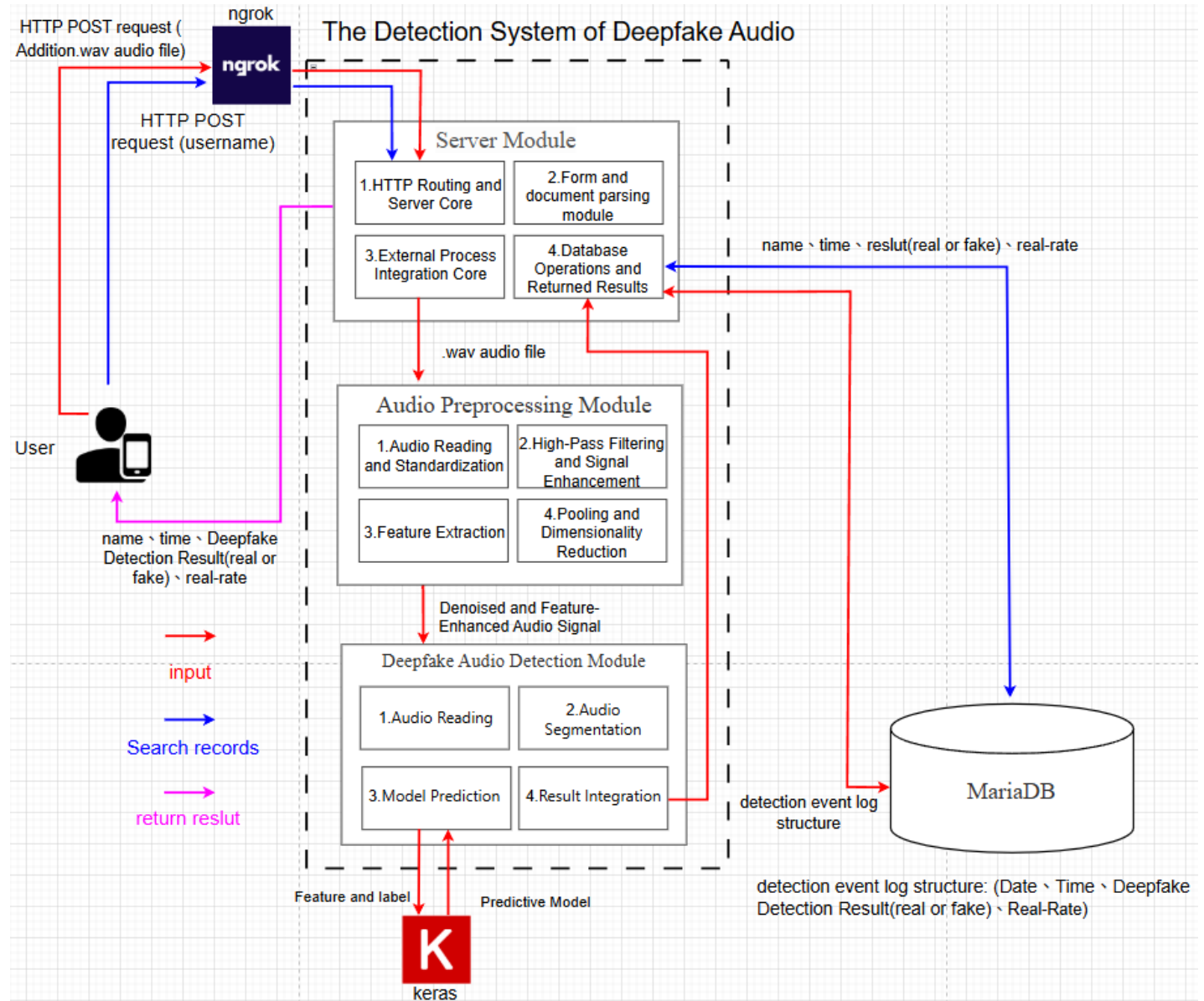
Purpose

- Develop a Deepfake voice detection system.
- Simple and quick way to identify deepfake audio.
- Reduce the success rate of scams.

Literature Review

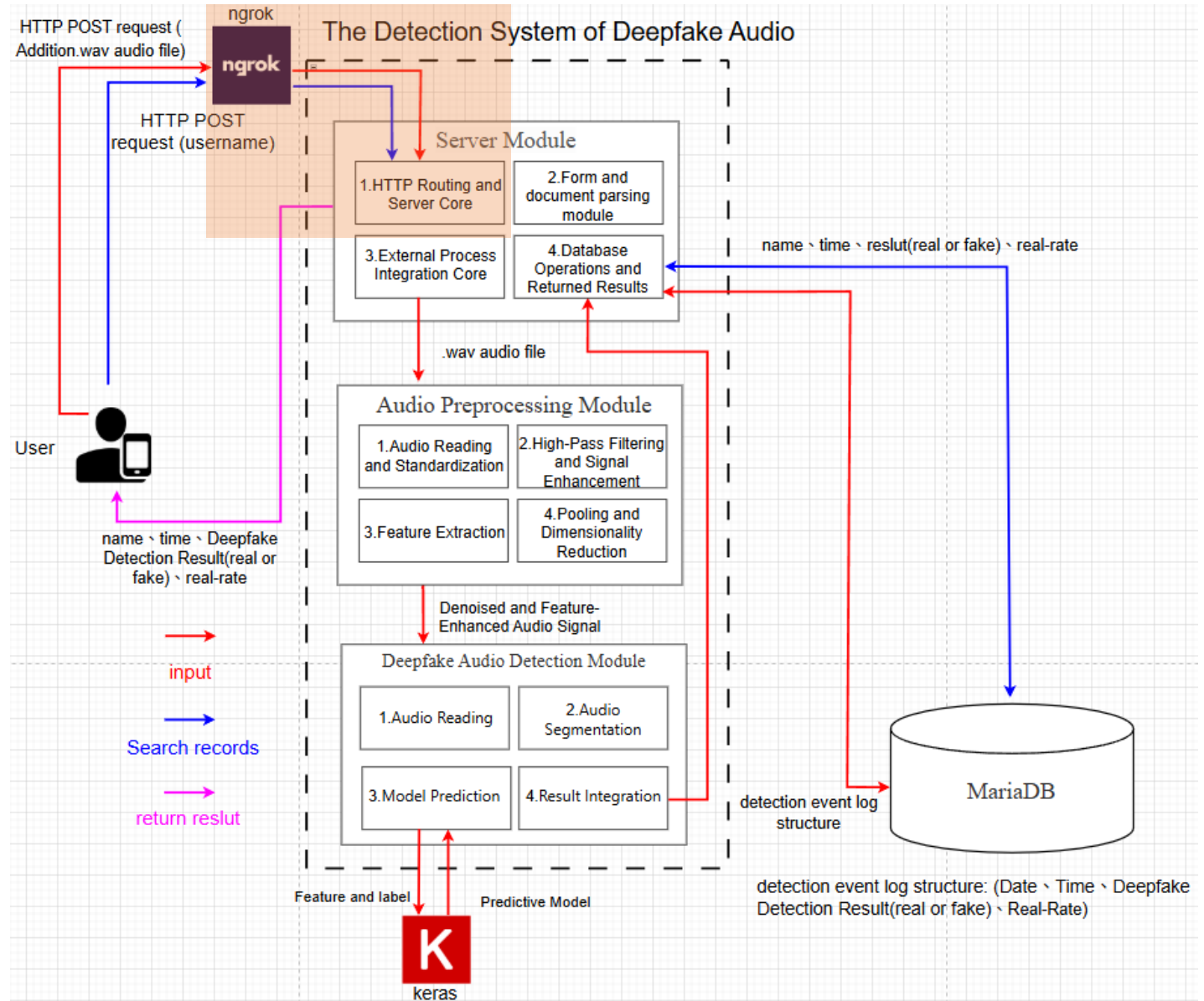
Name	Description
Deepfake	Deepfake are images, videos, or audio that have been edited or generated using artificial intelligence
Mel-Frequency Cepstral Coefficients (MFCC)	MFCC are a widely used feature extraction technique in the fields of speech processing, speech recognition, and music information retrieval
Spectrogram	Spectrogram is visual representation of the spectrum of frequencies of a signal as it varies with time.
High-pass filter (HPF)	In audio processing, HPF is tools to remove the lower-frequency content of an audio signal, allowing the higher frequencies to pass

Architecture



Architecture

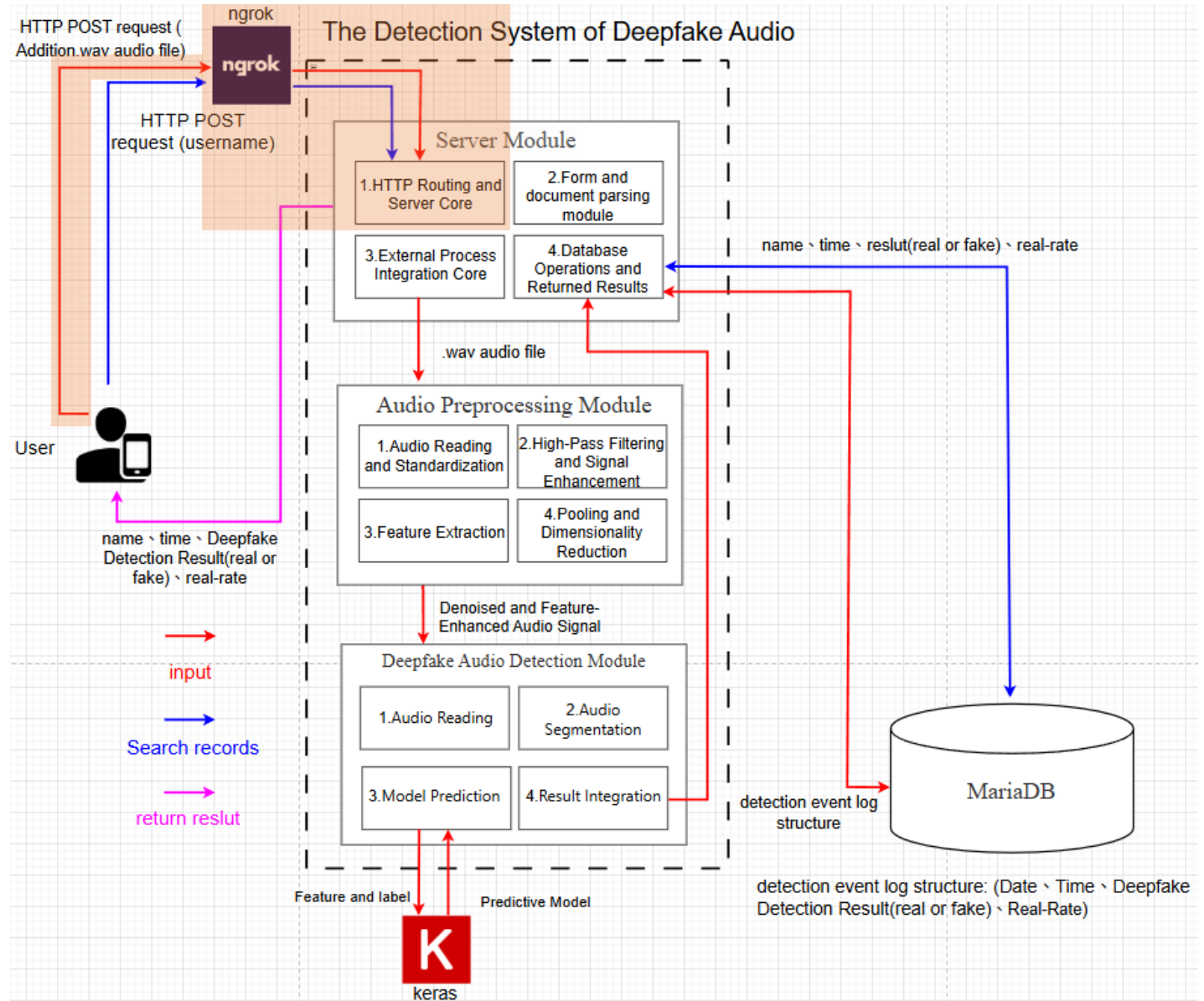
First, after the server starts, it establishes a secure TLS/SSL tunnel with ngrok.



Architecture

User uploads audio files

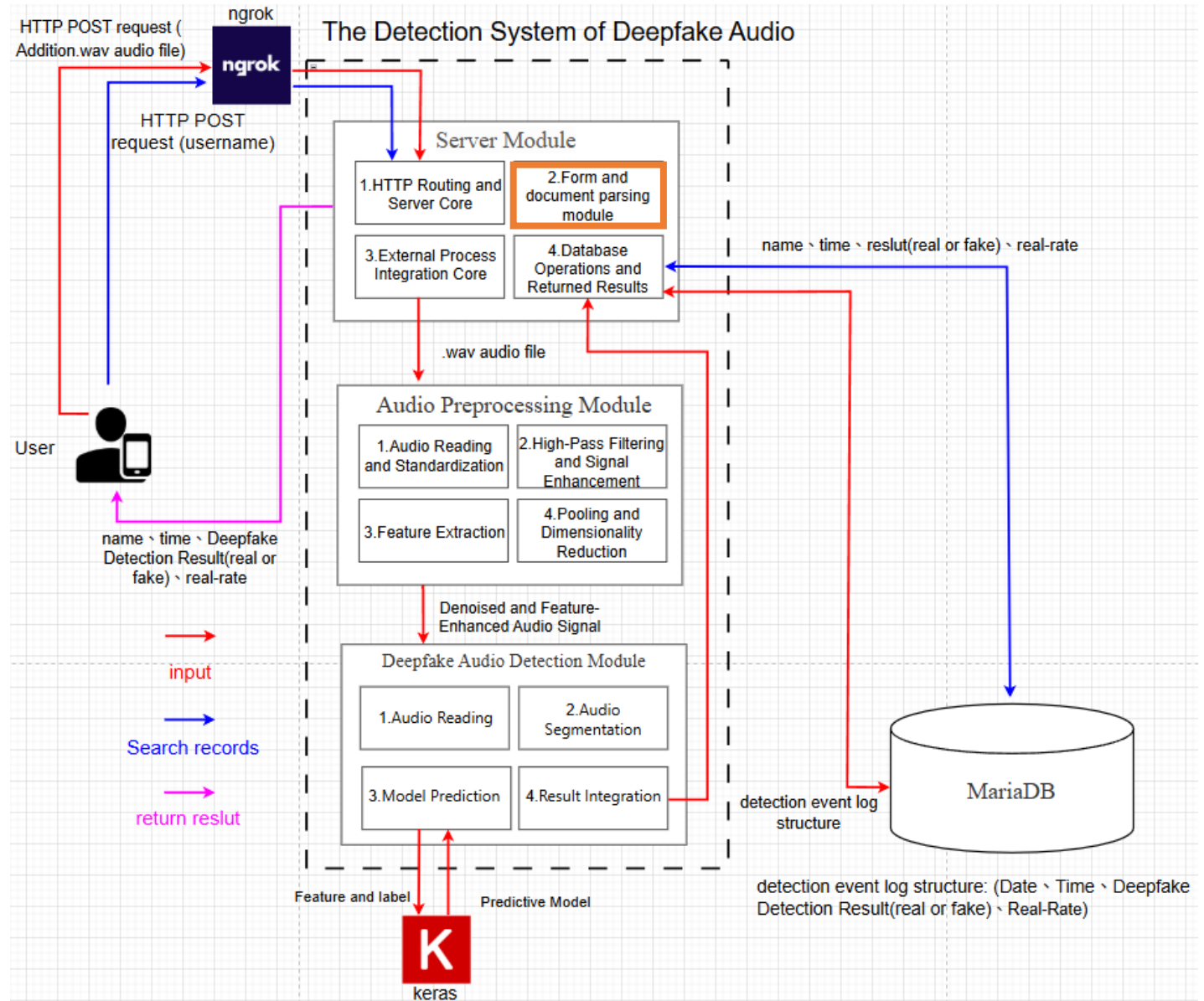
Server receives an HTTP request from the user containing an audio file.



Architecture

User uploads audio files

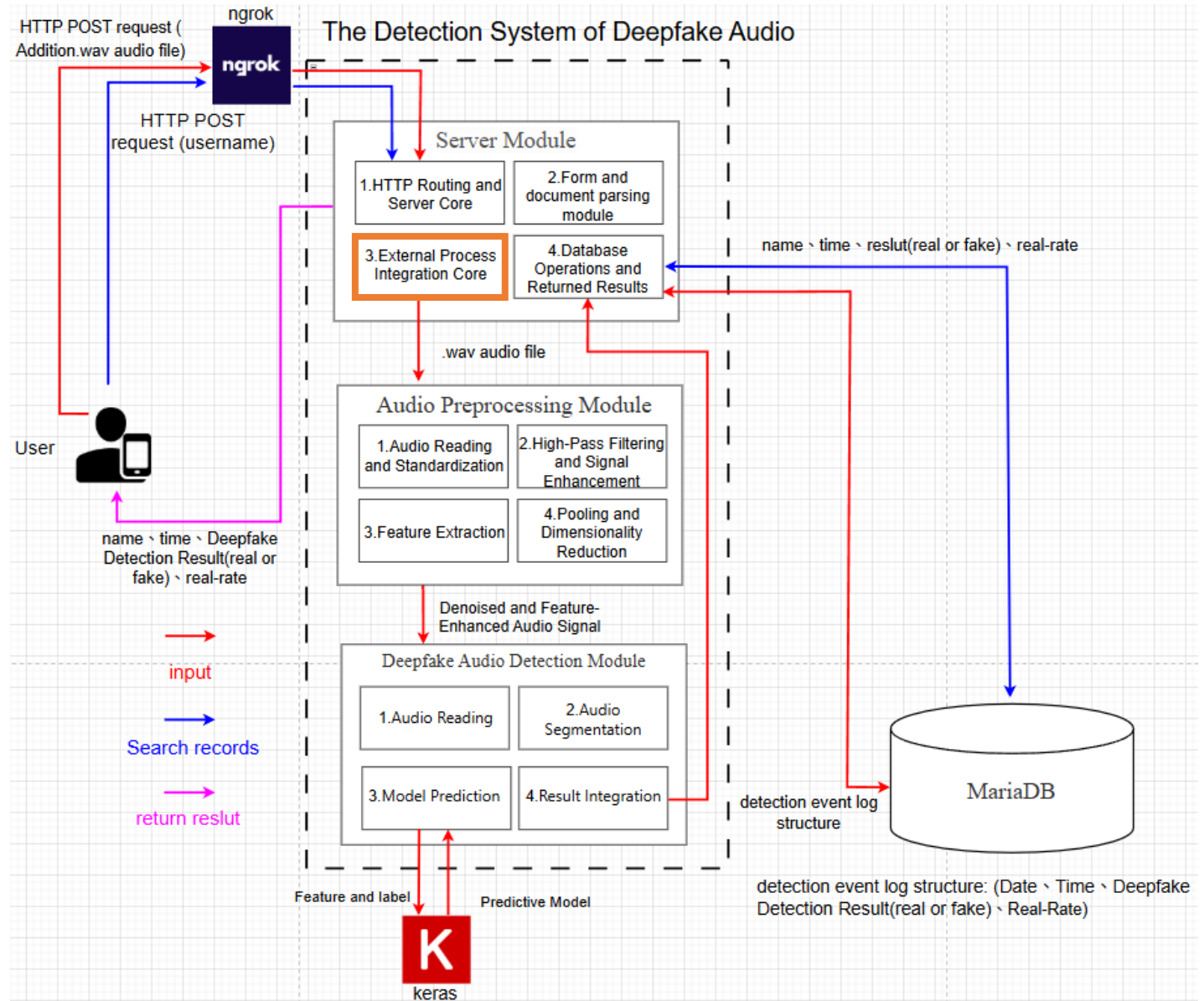
The form and file parsing module parses the HTTP stream, extracts multimedia files and form fields, and performs buffered write-to-disk I/O on the server side.



Architecture

User uploads audio files

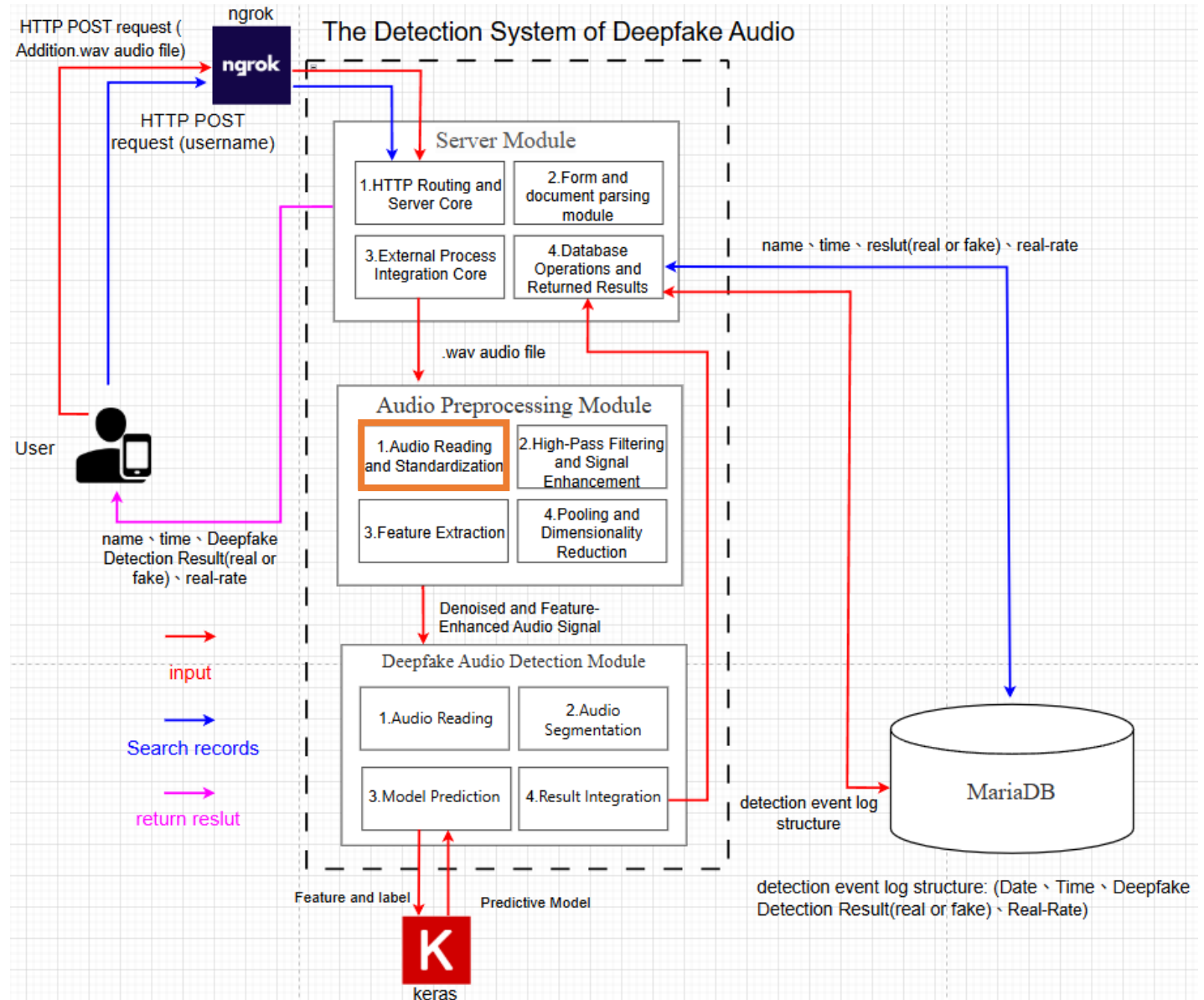
The server connects to other external programs and receives the returned results.



Architecture

User uploads audio files

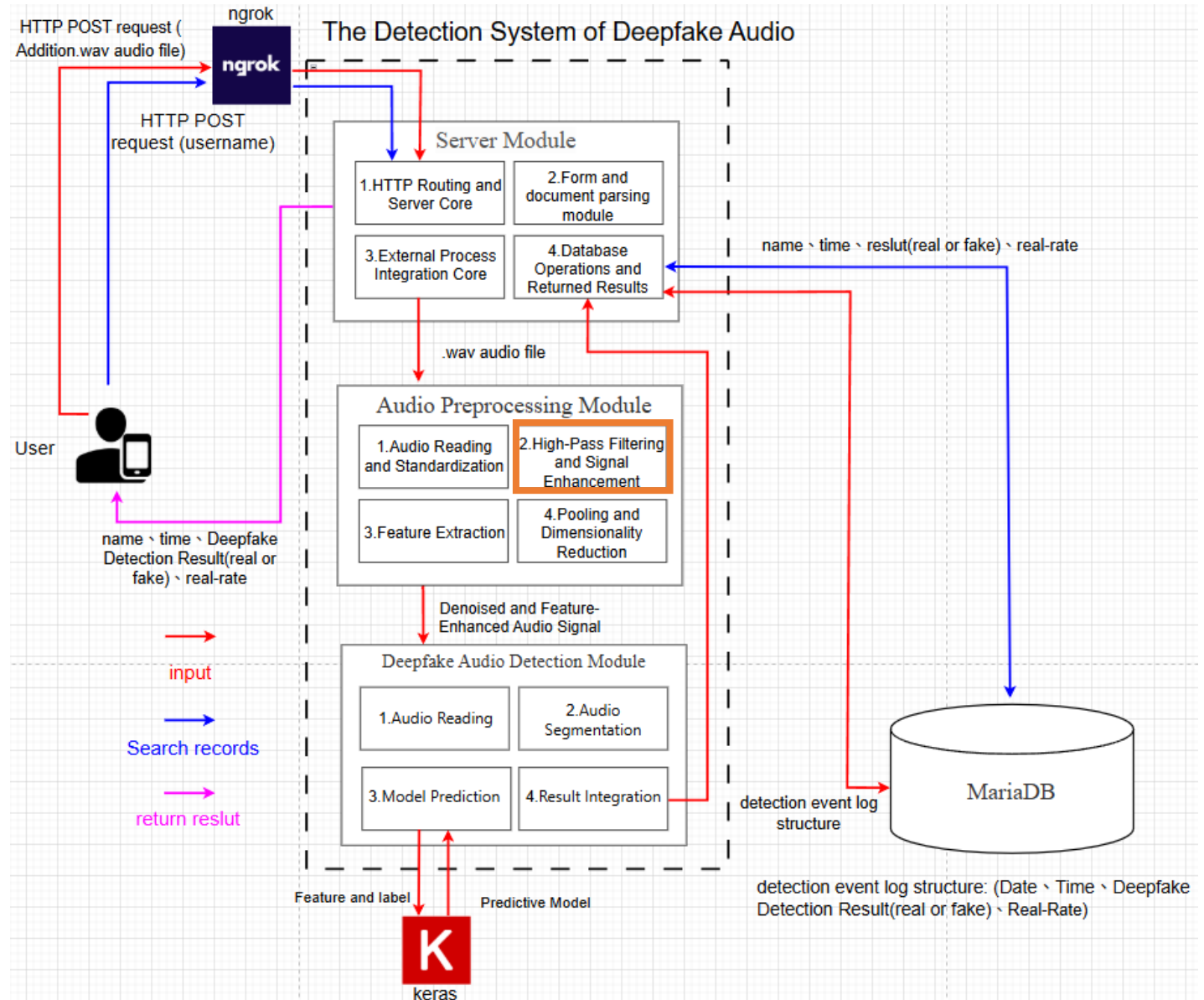
Read the uploaded audio file and ensure that its sampling rate and channel characteristics are completely consistent.



Architecture

User uploads audio files

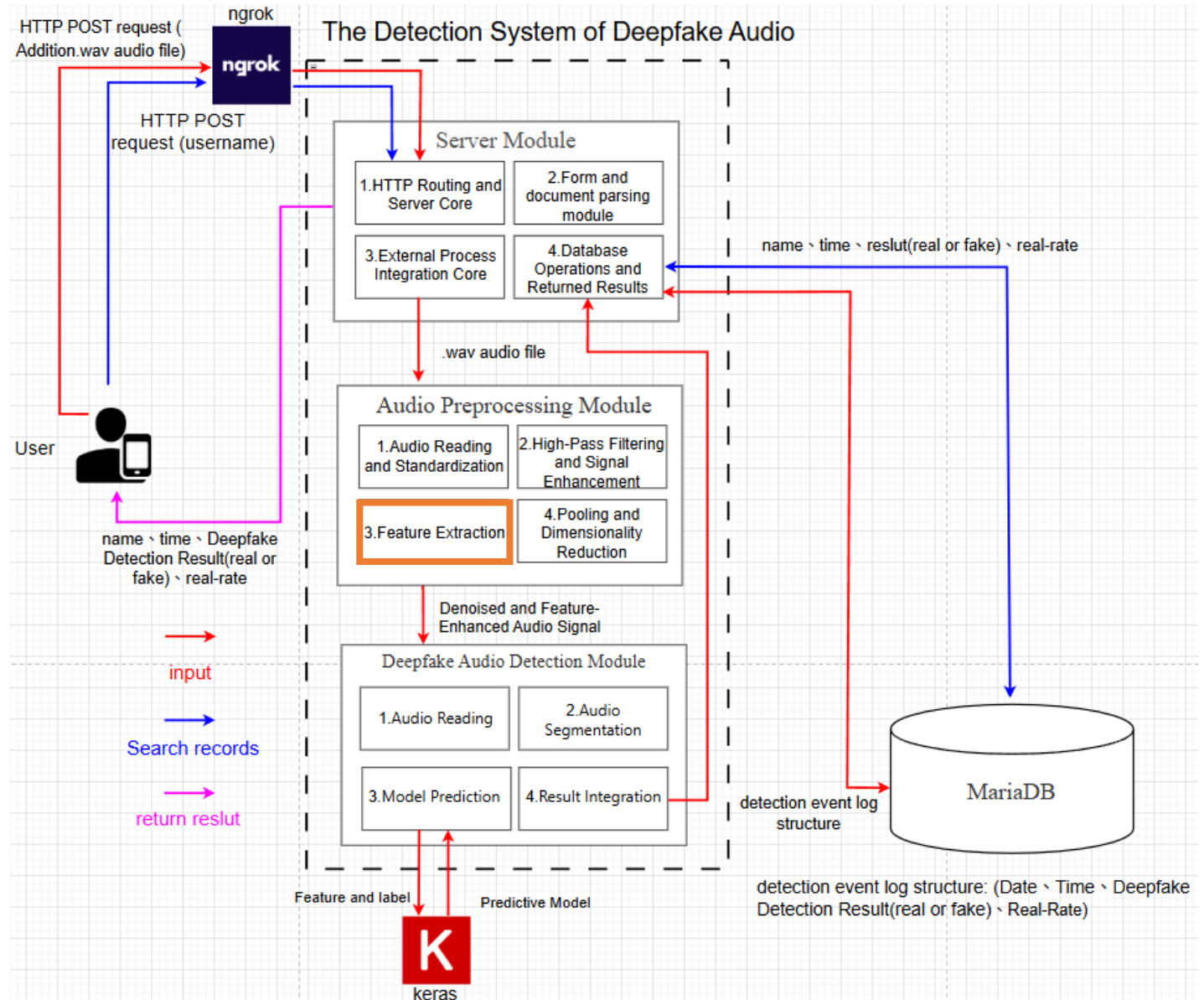
High-pass filters are used to filter out low-frequency noise in the environment and enhance voice signals.



Architecture

User uploads audio files

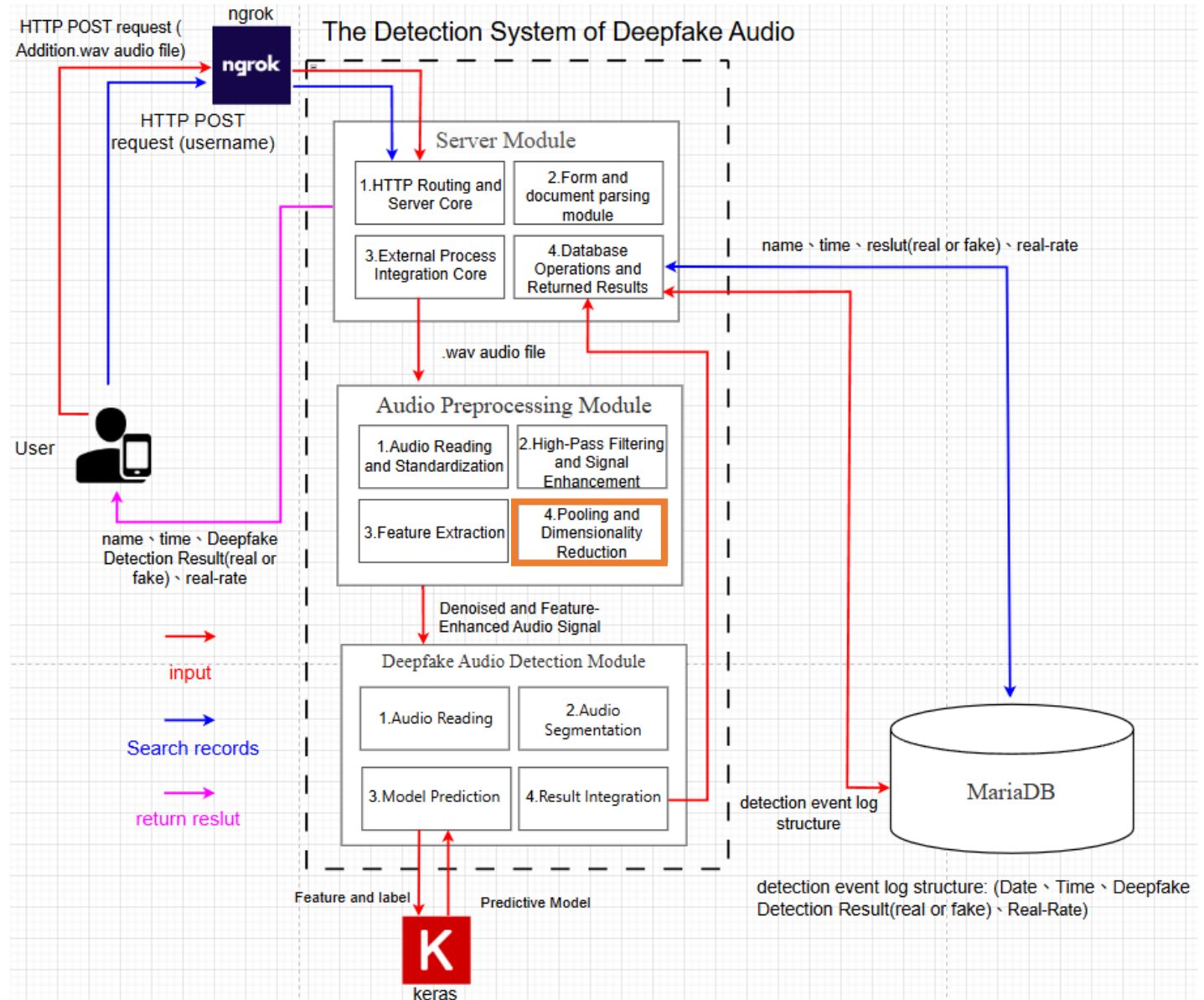
The enhanced speech signal is converted into acoustic features (e.g., speech spectrograms) suitable as input to a deep learning model.



Architecture

User uploads audio files

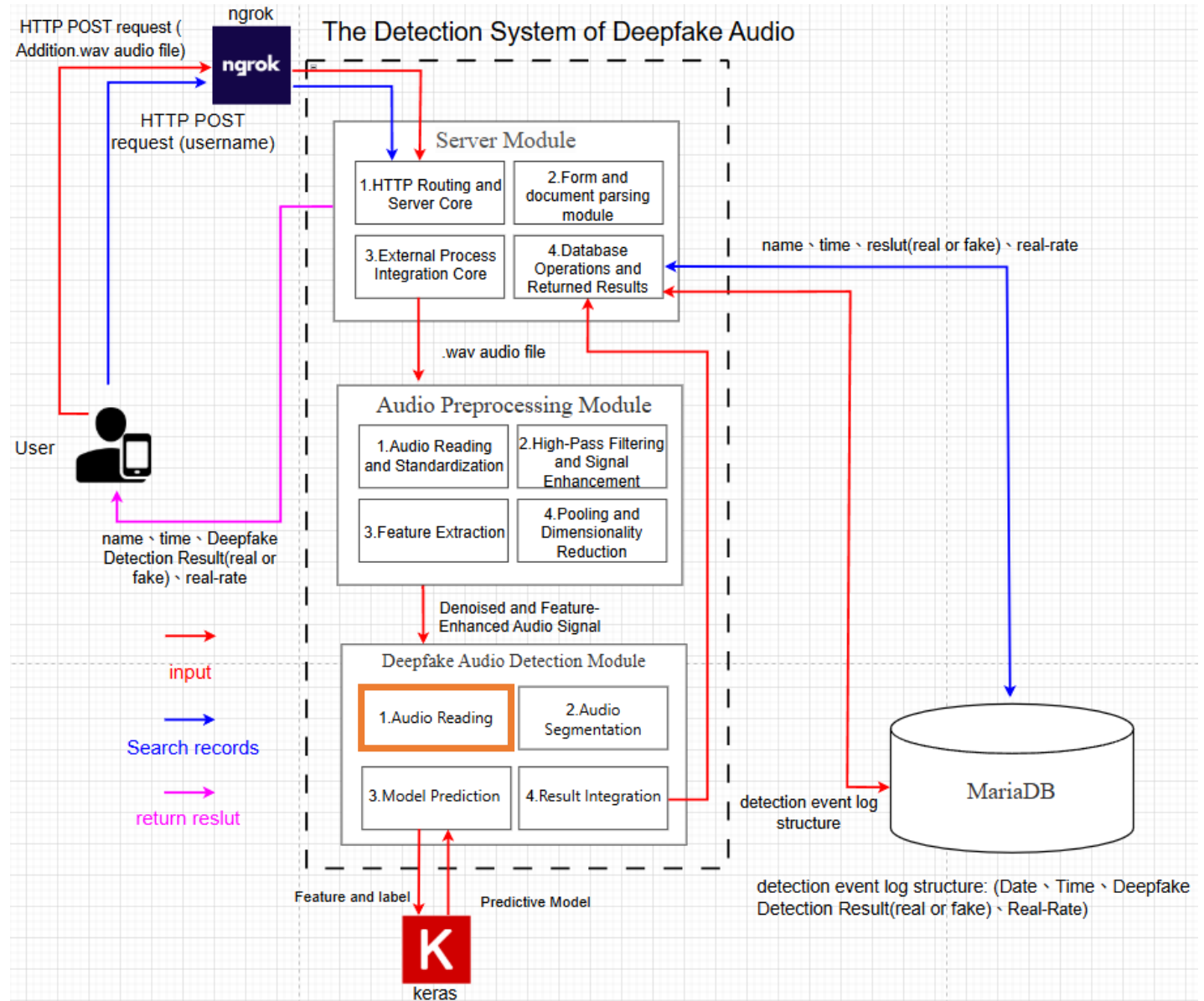
By using pooling techniques to simplify feature dimensions and extract key features, the final output is a "noise-reduced and feature-enhanced audio signal" that can be used in the next stage.



Architecture

User uploads audio files

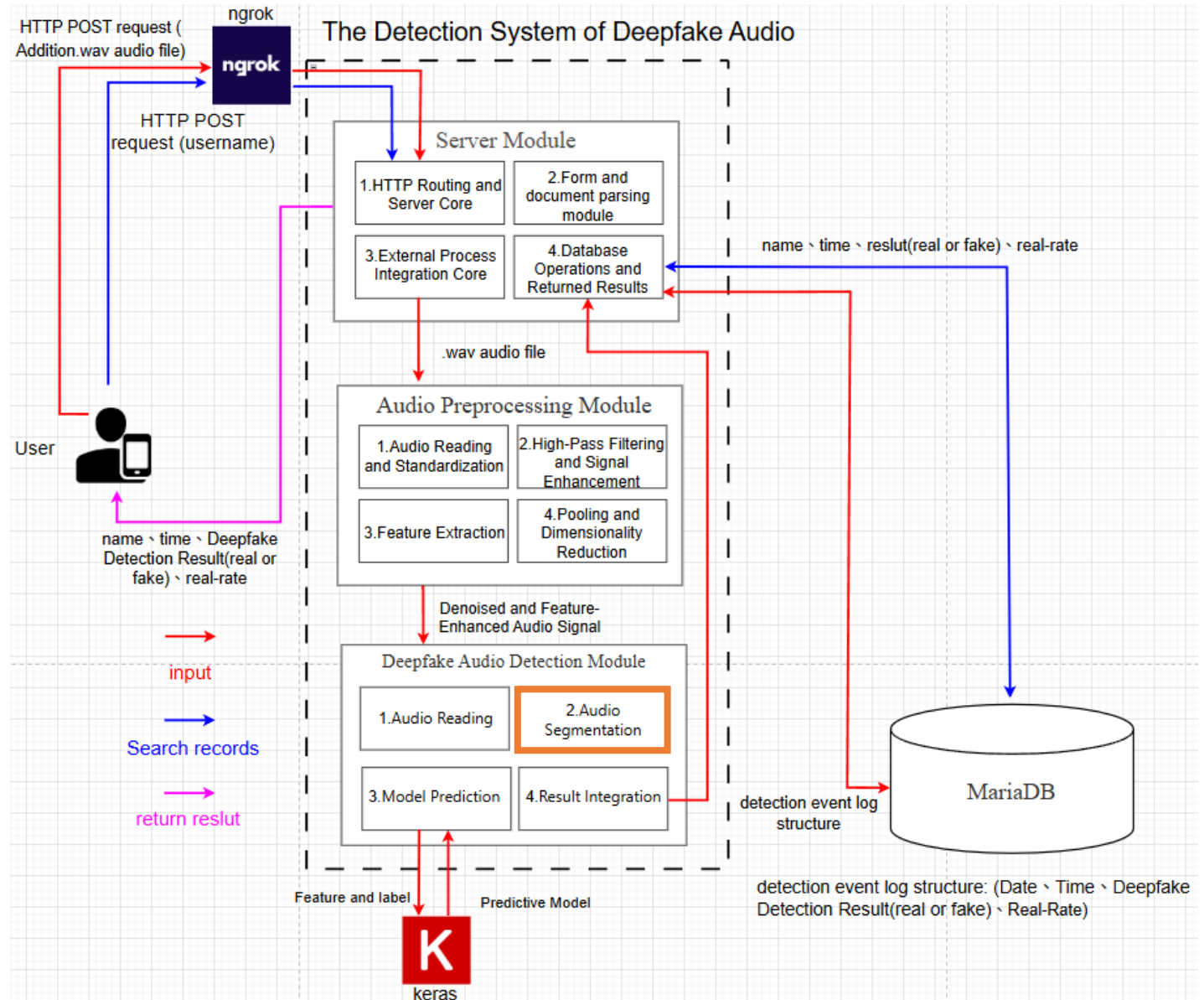
Receives enhanced feature signal output from the preprocessing module.



Architecture

User uploads audio files

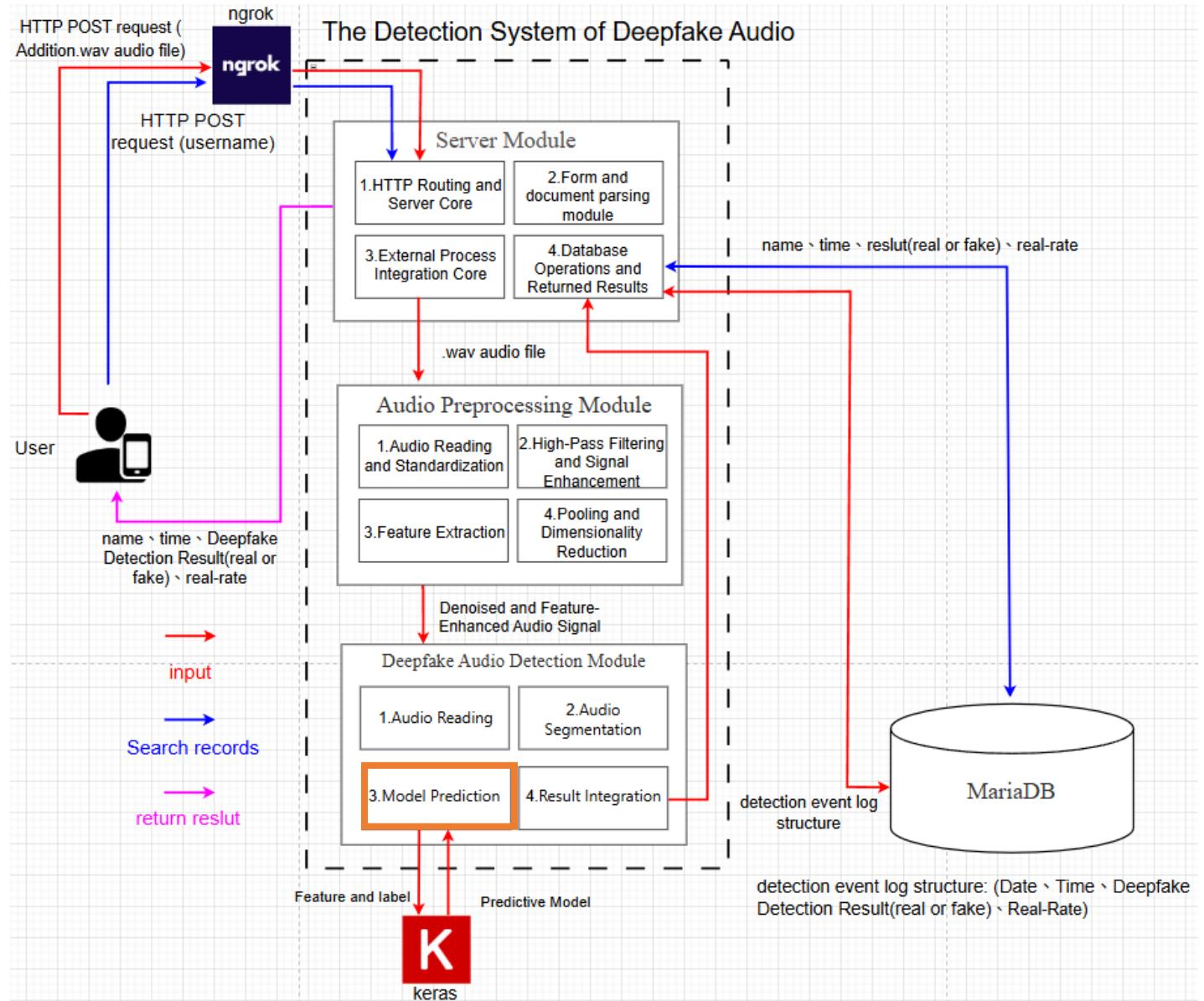
Longer speech segments are segmented into segments of specific time lengths so that the model can perform detailed analysis of local speech features.



Architecture

User uploads audio files

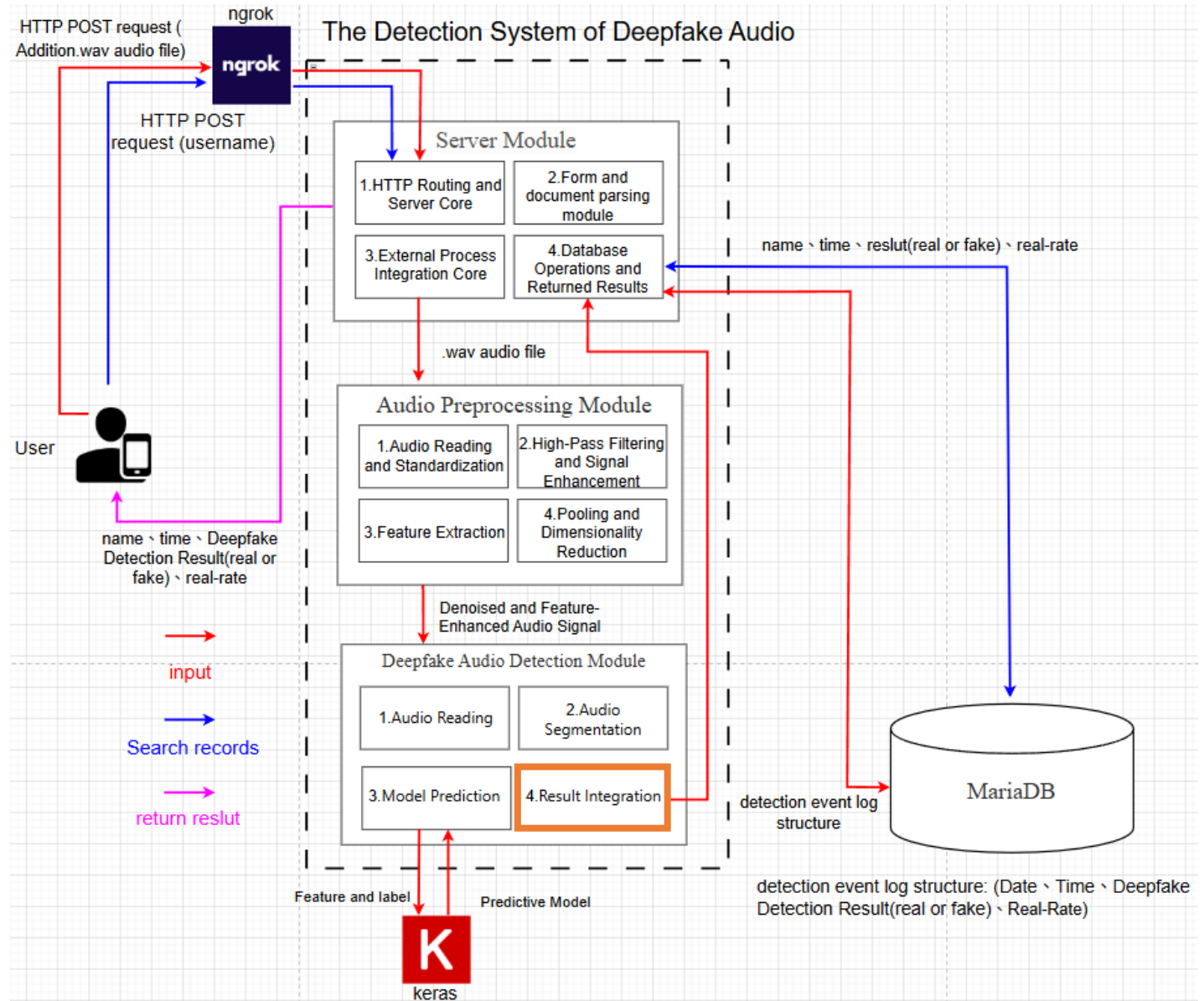
This step uses the external Keras deep learning API to train an artificial neural network (ANN) model, taking features as input and training it. then we put the segmented audio files as input into trained model for evaluation.



Architecture

User uploads audio files

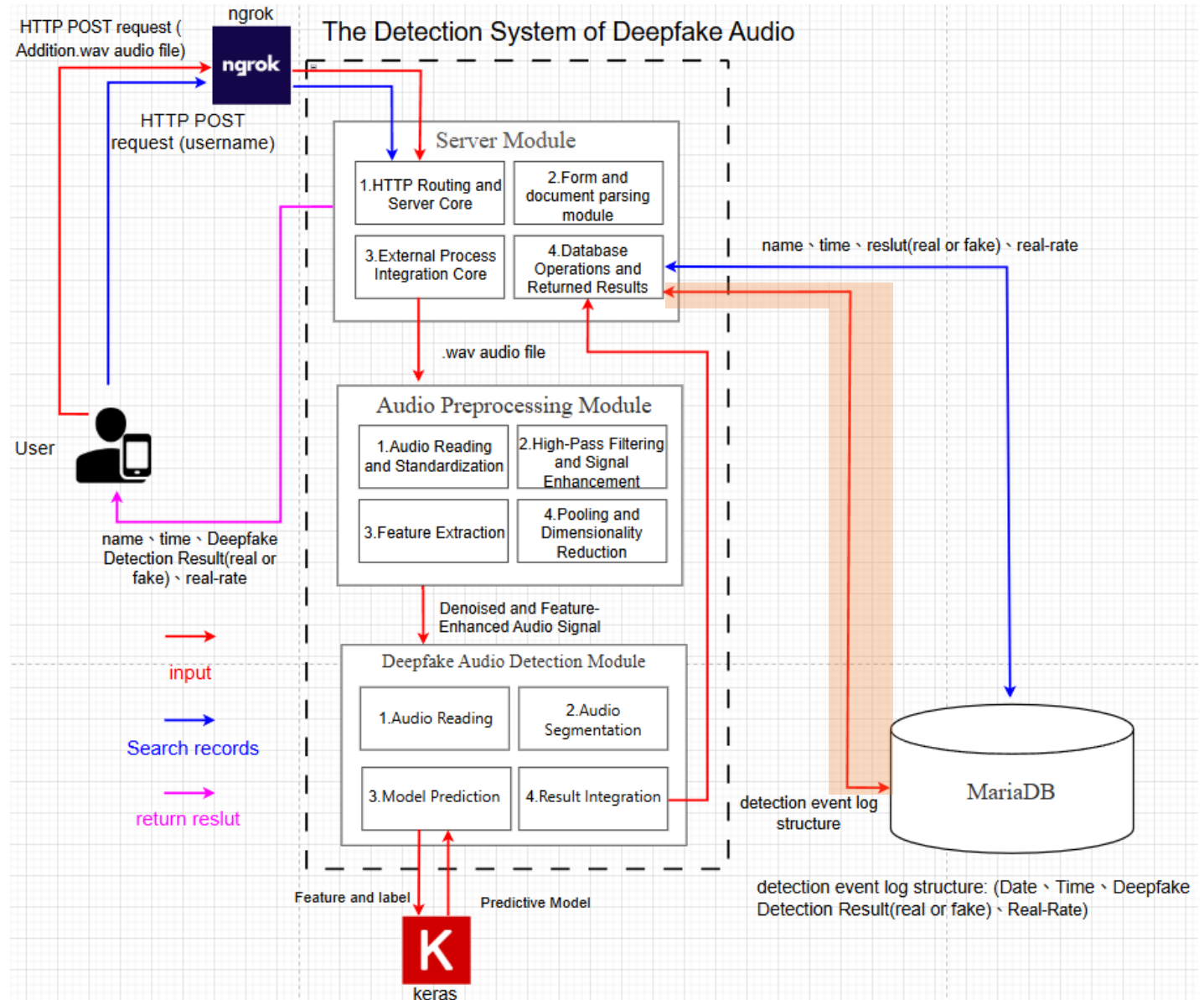
By integrating the predicted probabilities of each audio segment, the final judgment result is calculated and sent back to the server.



Architecture

User uploads audio files

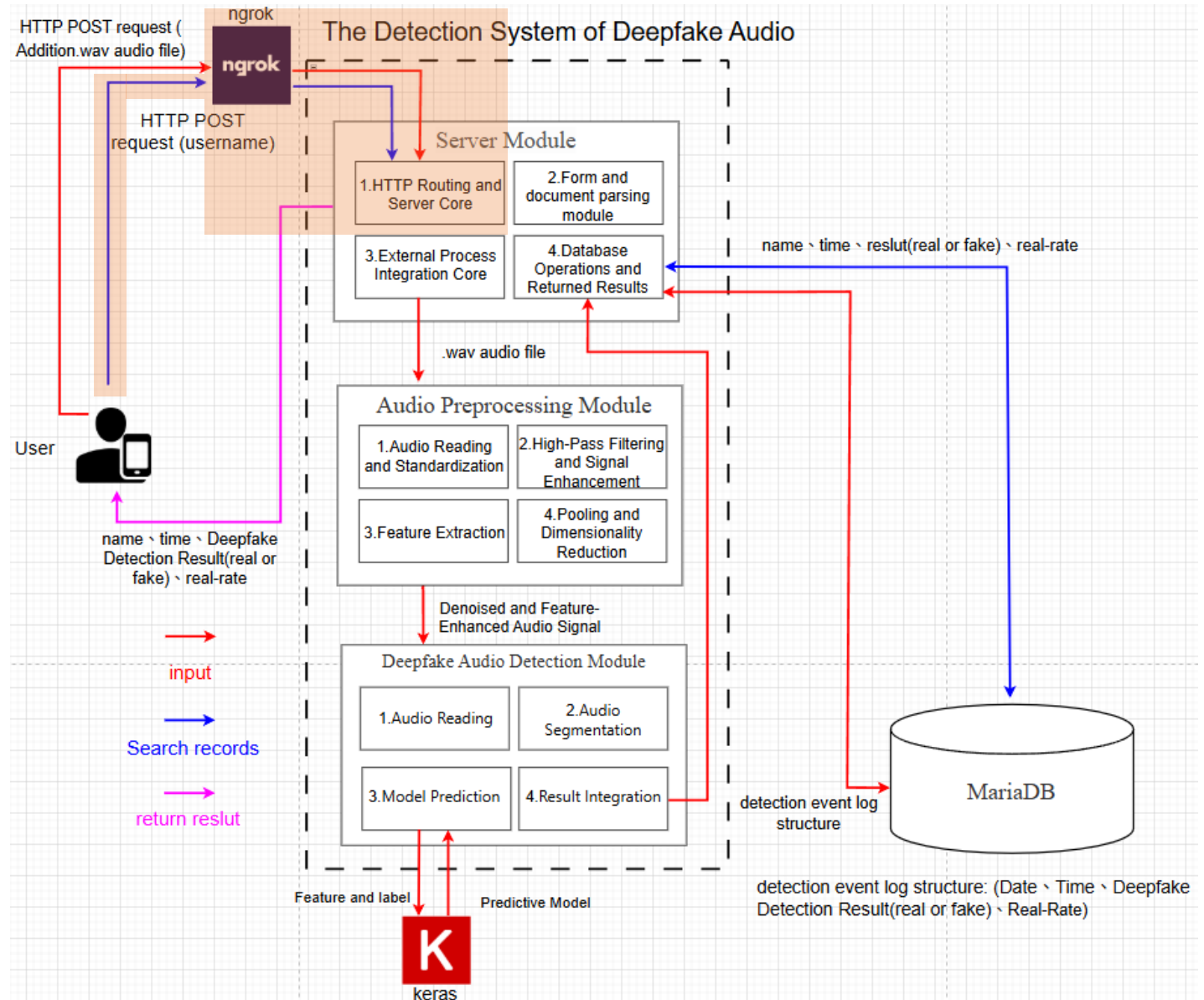
The server module returns the results and transmits them to the user and database for recording.



Architecture

Users only query historical records

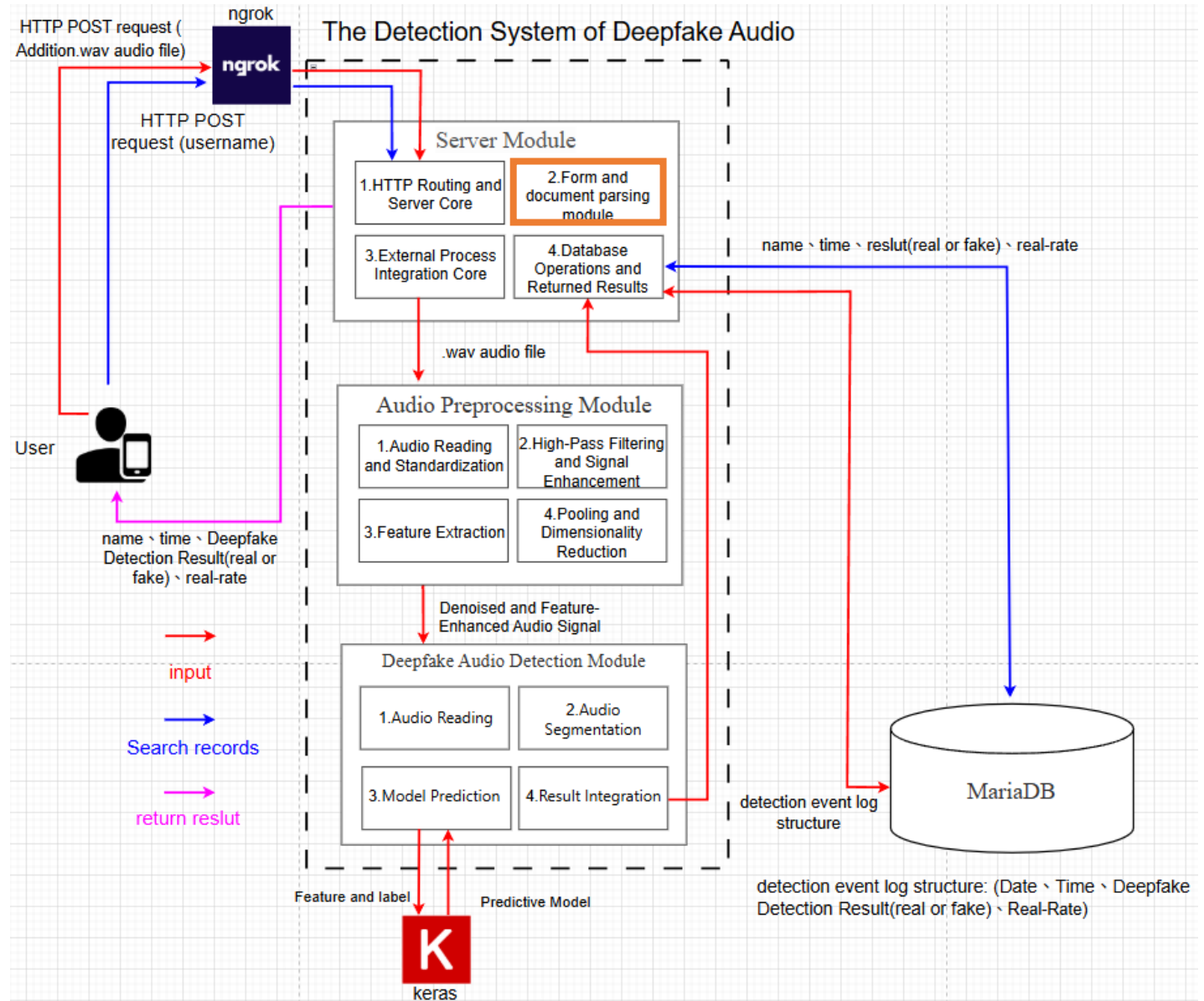
Server receives an HTTP request from the user containing an username.



Architecture

Users only query historical records

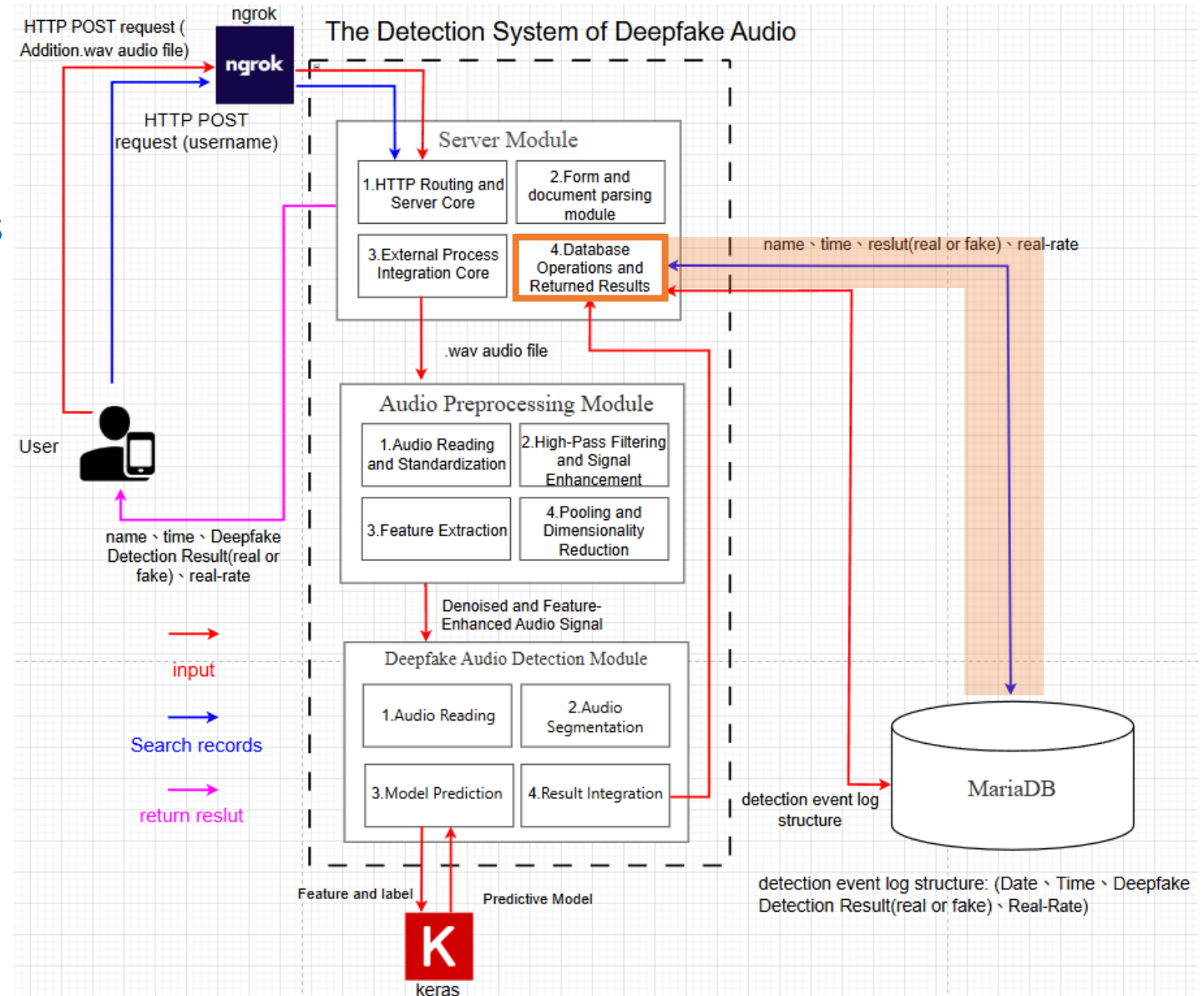
The form and file parsing module parses HTTP streams and extracts multimedia files and form fields.



Architecture

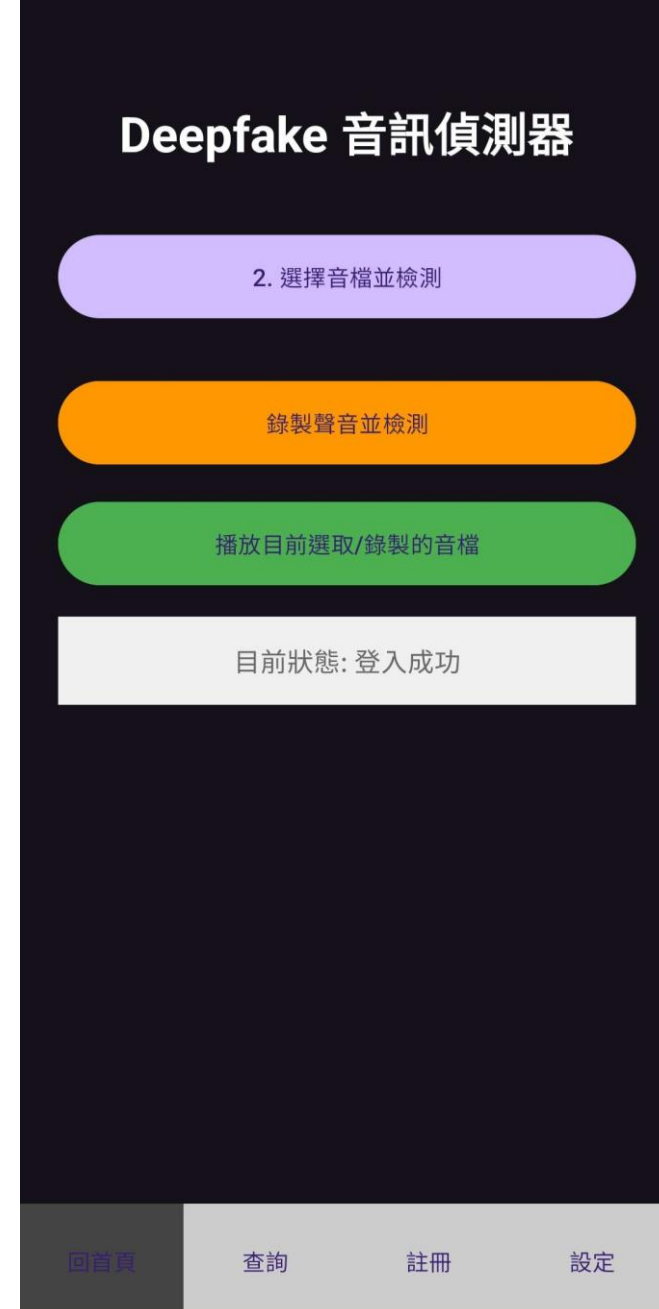
Users only query historical records

The server queries the database based on user information and returns the results.



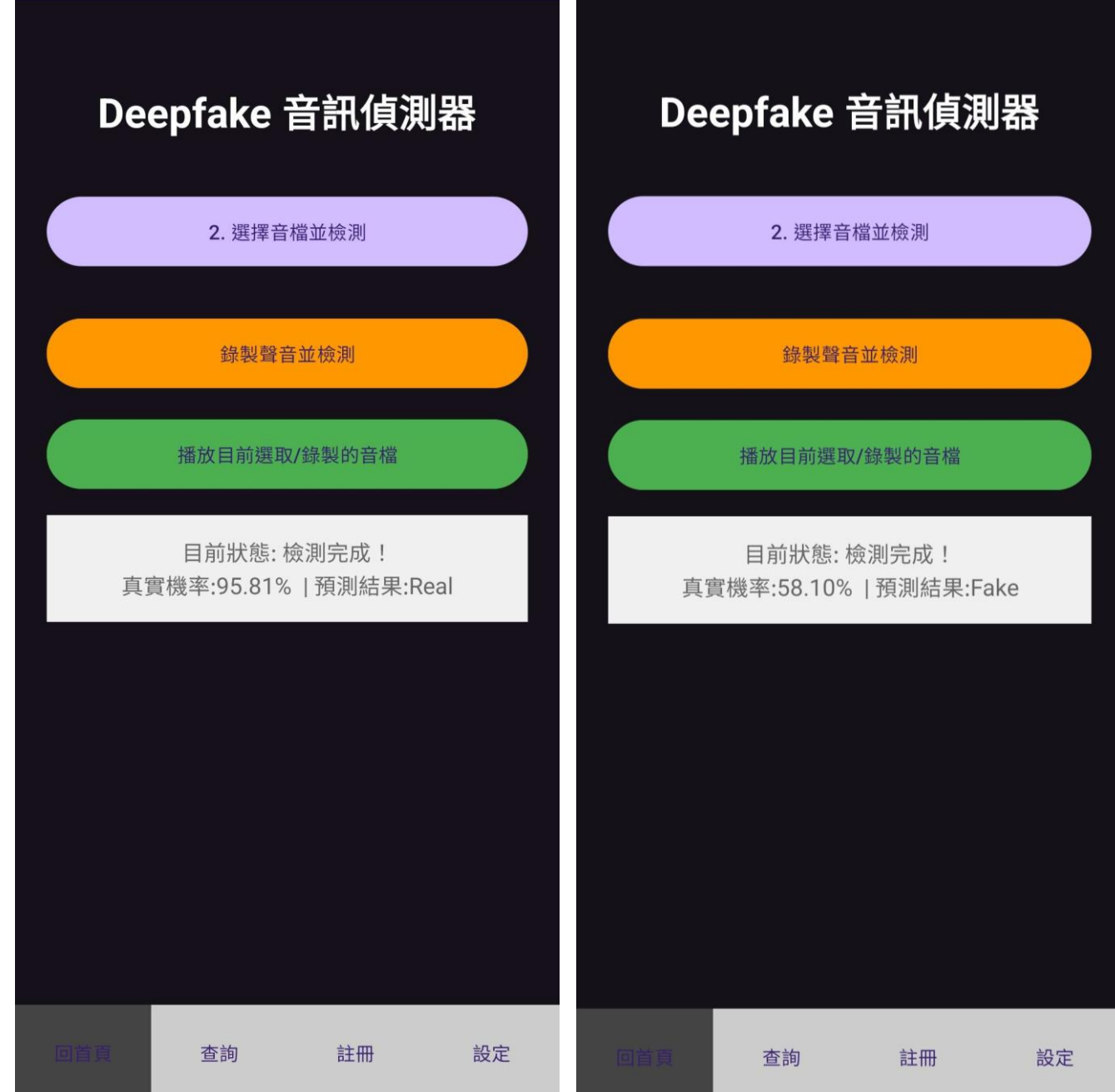
Experiment

- Use mobile app to upload audio file or recording audio.



Experiment

- Server will return the detection result, identifying whether the audio is authentic or spoofed.



Conclusion

- 1. With the development of artificial intelligence, deepfake voice recordings are becoming increasingly difficult to detect.
- 2. We developed a speech detection model with an accuracy rate of 93.65%.

Reference

- [1]"Deepfake." Wikipedia, Wikimedia Foundation, 17 May 2019 <https://en.wikipedia.org/wiki/Deepfake>
- [2]"Mel-Frequency Cepstrum." Wikipedia, 21 Dec 2019 https://en.wikipedia.org/wiki/Mel-frequency_cepstrum
- [3]"Spectrogram." Wikipedia Contributors, 17 Oct 2019 <https://en.wikipedia.org/wiki/Spectrogram>
- [4]"High-pass filter." Wikipedia Contributors, 22 Nov 2019 https://en.wikipedia.org/wiki/High-pass_filter

System Environment

System Development Environment

- Hardware: Personal computer
- Operating System: Windows 11 、 Android 16
- Programming Language: Python 、 Java script
- Editor: Android studio 、 Visual Studio Code

User Environment

- Hardware: smartphone
- Operating System: Android 16
- Operation Interfaces: smartphone